



A data-driven approach to determine the minimum number of spot samples for estimating methane and carbon dioxide emissions from Austrian dairy cows on commercial farms using GreenFeed

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ABSTRACT

Accurate estimation of methane (CH₄) and carbon dioxide (CO₂) emissions requires precise sampling methods due to the significant variability of these gases. This study, part of the project breed4green in Austria, investigated the variability of CH₄ and CO₂ and minimum number of spot samples required to accurately estimate CH₄ and CO₂ emissions from dairy cows on commercial farms. A total of 51,895 spot samples from 451 cows on 9 farms was analyzed; these samples were obtained using the GreenFeed system. Each farm participated in 2 GreenFeed measurement periods (period 1 and 2), each lasting 6 wk and separated by 3 to 6 mo. A total of 298 cows, each with at least 70 spot samples in either period 1 or period 2, were analyzed to determine the minimum number of spot samples. Two methods were used. In the first, forward sampling, subsets of increasing spot samples (the first 1, 10, 20, 30, 40, 50, 60, and 70 measurements per cow) were compared with the complete set of available measurements for the cows. In the second method, random sampling, the same number of spot samples was selected randomly from all available visits for each cow and this procedure was repeated 100 times to account for sampling variability. The minimum number of spot samples required was determined when the Pearson correlation between the arithmetic mean of the subset and that of all available measurements reached a value of 0.90. Mean CH₄ and CO₂ emissions were 430 g/d and 13,309 g/d, respectively. The CV for diurnal variability ranged from 4.6% to 12.0% for CH₄ and from 2.7% to 6.9% for CO₂, indicating low to moderate diurnal variability. Day-to-day variability was low, with CV between 3.7% and 8.7% for CH₄ and between 2.5% and 5.5% for CO₂. Substantial within-cow variation was

observed for CH₄ (mean CV of 22.1% in period 1 and 22.9% in period 2), whereas CO₂ was more stable (mean CV of 11.8% in period 1 and 11.5% in period 2). When using forward sampling, at least 19 spot samples were required for accurate CH₄ estimates and only 13 for CO₂, with a mean of 6.5 and 4.3 d needed to obtain these measurements, respectively. When using random sampling, only 8 spot samples for CH₄ and 5 spot samples for CO₂ were sufficient to meet the accuracy criteria. For future analyses of this dataset, we will rely on the results of forward sampling. Forward sampling is based on the actual sequence of visits made by each cow to the GreenFeed and therefore offers a more conservative and realistic approach compared with random sampling. Overall, the results highlight the dynamic nature of CH₄ emissions, which are influenced by factors such as feeding patterns and rumen fluctuations and suggest that more spot samples are needed for reliable estimates of CH₄ than for CO₂.

Key words: GreenFeed, methane and carbon dioxide, spot samples, commercial farm

INTRODUCTION

Ruminant livestock are essential for global food security because they can convert human-inedible roughage and agricultural byproducts into high-value animal proteins such as meat and milk. However, a natural byproduct of their digestive process is enteric methane (CH₄), a potent GHG. In Austria, the decrease in the number of dairy cows and the increase in milk yield have contributed to a reduction of GHG emissions during the past few decades (Hörtenhuber et al., 2022, 2023). Even more progress may potentially be made through direct selection for reduced CH₄ emissions because significant differences in CH₄ production have been observed between individual animals (Heringstad et al., 2023; van Breukelen et al., 2023; Chipondoro, 2024). de Haas et al. (2021) showed that direct selection on CH₄ has the

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

potential to reduce emissions substantially. By putting economic weight on CH₄ in the breeding goal, selective breeding can reduce CH₄ emissions by 24% in 2050. To enable direct selection based on CH₄ emission traits, accurate measurements are essential.

Technologies such as GreenFeed (C-Lock Inc., Rapid City, SD) are used to measure CH₄ and CO₂ emissions in dairy cattle. This system captures spot samples, typically lasting 2 to 4 min. The measurements are taken at various times throughout the day over multiple days while cows are fed small amounts of concentrate. To accurately determine CH₄ and CO₂ emissions for an individual animal, a sufficient number of spot samples is required because CH₄ and CO₂ show diurnal variation (Manafiazar et al., 2017; Islam et al., 2023; Raynor et al., 2025). Methane production in cows is mainly driven by rumen fermentation of carbohydrates, with emissions peaking in the second or third hour after feeding (Brask et al., 2015; Olijhoek et al., 2016; van Lingen et al., 2017).

Collection procedures of spot samples using the GreenFeed system differ across studies. Some previous studies, including those by Ma et al. (2024), Smith et al. (2024), and Pressman et al. (2025), applied fixed sampling schemes to obtain representative emission values. For example, Smith et al. (2024) measured CH₄ 15 times per cow, with cows being measured at 0800, 1200, and 1600 h from Monday to Friday during the measurement week, with each cow being assessed for at least 10 min in the same order. In a recent study, Lee et al. (2022) determined that in a fixed sampling schedule at least 8 samples (every 3 h in a 24-h cycle) were required to estimate daily CH₄ and CO₂ production. In contrast, other studies (Dressler et al., 2023; Koning et al., 2024; MacNeil and Waterman, 2025) collected data through voluntary visits to the GreenFeed. Various analyses of GreenFeed data have led to a range of recommendations, with the number of samples required per animal ranging from 10 to 40 over a period of 7 to 45 d (Manafiazar et al., 2017; Koning et al., 2024; MacNeil and Waterman, 2025). As noted by Manafiazar et al. (2017), studies with less variable CH₄ emissions patterns may require shorter and less intensive sampling, and studies with more variable CH₄ patterns will require longer and more intensive sampling. One factor that influences these requirements is the production system. Koning et al. (2024), for example, found that pasture systems require more spot sample measurements with GreenFeed than indoor housing systems to obtain reliable measurements of enteric CH₄ production in dairy cattle.

Most existing research has been conducted under controlled conditions at research facilities, which may restrict its applicability to commercial farming settings. In this study, we aimed to assess variability in CH₄ and CO₂ and to determine the minimum number of spot samples

required to accurately estimate CH₄ and CO₂ emissions from dairy cows on commercial farms with indoor feeding systems in Austria. We hypothesized that a larger number of spot samples would be required to accurately estimate CH₄ and CO₂ emissions in commercial farms because feeding and management practices are subject to greater variability compared with research facilities. This study, which was carried out as part of the project breed4green, is an important first step toward developing a breeding value estimation for reducing CH₄ emissions in Austrian dairy cows.

MATERIALS AND METHODS

Experimental Design

Data from the project breed4green (Linke et al., 2025) were available from December 14, 2023, to June 24, 2025, from 9 commercial farms with herd sizes ranging from 33 to 62 dairy cows. An overview of the 9 farms is presented in Table 1. All farms kept Fleckvieh or Brown Swiss dairy cows in freestall barns equipped with sensor technology and either automatic milking systems or milking parlors with automatic milk meters. The cows were fed indoors, with no grazing and had ad libitum access to drinking water. The average milk yield per cow and year on the farms ranged from 6,641 to 10,887 kg. All farms fed grass silage and corn silage, with 1 farm feeding grass silage only. In addition, the cows were fed either TMR or a partial mixed ration, the latter being supplemented with individual concentrate feeding provided via automatic feeding stations.

An overview of the data recording on commercial farms is given in Figure 1. The GreenFeed system (C-Lock Inc., Rapid City, SD) was used to measure CH₄ and CO₂ on farms over a 6-wk period (Figure 1), with no training period. Body weight was recorded over the 6-wk period using the Tru-Test Walk Over Weighing system from Datamars (Australia Pty Ltd., Banyo, QLD, Australia). The GreenFeed unit was set up in the holding area or hoof trimming area, which was always accessible to the animals. The scales were placed in an aisle after the milking system. The cows were weighed separately from the GreenFeed visit each time after milking. One GreenFeed unit was installed on each farm. When using GreenFeed in confinement the general recommendation is 20 to 40 animals per unit (Harrison, 2025). To account for cows not using the GreenFeed and for cows unavailable during the dry period, farms with slightly larger herd sizes were primarily selected to ensure sufficient measurements. Weekly milk recording, bulk tank milk sampling, and feed analyses were conducted during the 6-wk GreenFeed measurement period. After the initial 6-wk period, the GreenFeed and scales were cleaned, disinfected and

Table 1. Overview of the commercial farms, including main breed (FV = Fleckvieh, BS = Brown Swiss), number of cows, milk yield, feed ration composition and GreenFeed measurement periods

Farm	Main breed	Number of cows ¹	Milk yield ² (kg)	Feed ration composition	GreenFeed period 1 ³		GreenFeed period 2 ³	
					Start	End	Start	End
A	FV	62	10,887	Grass silage + corn silage + barley straw + concentrate	12/14/2023	1/31/2024	6/6/2024	7/15/2024
B	FV	61	9,593	Grass silage + corn silage + barley straw + concentrate	2/1/2024	3/13/2024	7/16/2024	8/27/2024
C	FV	39	10,627	Grass silage + barley straw + concentrate	3/14/2024	4/22/2024	8/28/2024	10/14/2024
D	FV	42	10,238	Grass silage + corn silage + hay + concentrate	4/4/2024	5/14/2024	8/22/2024	10/9/2024
E	FV	51	10,662	Grass silage + corn silage + hay + concentrate	4/23/2024	6/5/2024	11/28/2024	1/14/2025
F	FV	33	9,609	Grass silage + corn silage + hay + concentrate	5/15/2024	7/3/2024	1/17/2025	3/6/2025
G	BS	50	6,641	Grass silage + corn silage + concentrate	7/4/2024	8/21/2024	11/21/2024	1/8/2025
H	BS	58	6,724	Grass silage + corn silage + hay + concentrate	10/10/2024	11/20/2024	3/7/2025	5/9/2025
I	FV	55	10,629	Grass silage + corn silage + green rye + concentrate	10/15/2024	11/27/2024	5/15/2025	6/24/2025

¹Average number of cows in 2024.²Average milk year performance per cow in 2024.³Dates are given in month/day/year format.

transported to the next farm for measurement. Because 2 GreenFeed units and 2 scales were available for the project, data collection took place simultaneously on 2 farms. A second 6-wk measurement period took place 3 to 6 mo after the first. During the 2 GreenFeed measurement periods (Table 1), the feed ration composition was kept as consistent as possible. The GreenFeed measurements were mostly taken on the same cows in both periods, although the cows were at different stages of lactation. Throughout the whole 15-mo study period on each farm, additional data were collected, including BCS, lameness, veterinary diagnoses, hoof trimming data, feed ration composition, prophylactic measures (e.g., ketosis and milk fever prevention or treatment) and results of blood BHB tests. All dairy cows with GreenFeed measurements were genotyped. These additional data were not used in the present study but will be considered in future analyses. Farms participating in this project were not allowed to use feed additives specifically aimed at reducing CH₄ emissions.

Gas Emission Measurements

The GreenFeed is a feeding station for measuring CH₄ and CO₂ emissions of individual animals using radiofrequency identification (RFID) tags. The system encourages animals to visit by providing small amounts of pelleted feed. As the animal eats, a fan system draws air from around its muzzle into an intake manifold, which directs the airflow through the system. A subsample of

this air is then analyzed by nondispersive infrared sensors, which measure CH₄ and CO₂ concentrations at 1-s intervals. A detailed description of how CH₄ and CO₂ production is measured by the GreenFeed is provided by Hammond et al. (2015) and Huhtanen et al. (2015).

Maintenance and Settings

A CO₂ recovery test was carried out manually every 3 wk. A known amount of CO₂ gas was released in the GreenFeed units to check the recovery of expired gases. The average recovery of CO₂ of both units combined was 97.6%. The autocalibration was performed regularly using a zero (20% O₂ and 80% N₂; Linde Gas Benelux B.V.) and a span gas (0.05%–0.15% CH₄ and 0.5%–1.0% CO₂, 21% O₂, 0.01% H₂, and pure N₂; Linde Gas Benelux B.V.). The average calibration coefficients for CH₄ and CO₂ were 1.04 and 5.49, respectively. The animals were allowed to visit the GreenFeed up to 5 times a day, with a minimum interval of 5 h between visits to ensure that visits were evenly distributed throughout the day. During each visit, they received a maximum of 8 drops of pelleted concentrate feed, with each drop being given 30-s apart, resulting in a visit duration of approximately 4 min. The average weight of the cup drops was recorded at the start of each GreenFeed period on each farm. The average cup drop weight across all farms was 40 g, with weights ranging from 37 to 44 g. This resulted in a maximum additional concentrate intake of 1.48 kg to 1.76 kg per cow per

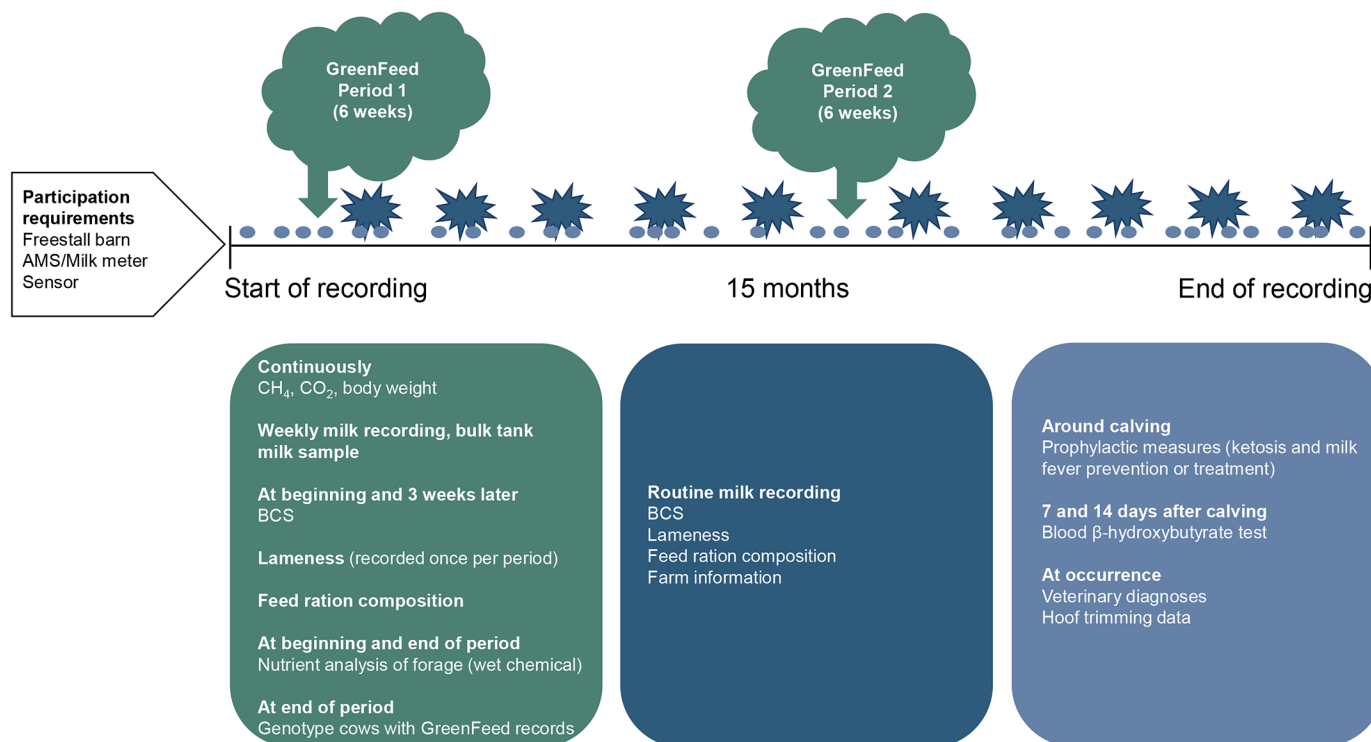


Figure 1. Overview of data recording on commercial farms. AMS = automatic milking system.

day. Animals that were not relevant to the project (e.g., Holstein) were excluded at the beginning of each GreenFeed measurement period and did not receive concentrate feed. After 3 wk of recording, cows with frequent visits (at least 100 visits of 2.5 min) were identified and no longer received concentrate feed. As a result, these animals visited the GreenFeed less frequently or for shorter periods of time. This gave other animals with fewer or no measurements more frequent access.

Multiple RFID Tags per Animal

Seven farms used 2 RFID tags per animal, one for the milking system and another electronic ear tag, which is required by law in Austria. The GreenFeed RFID reader alternated between reading the tags, which incorrectly suggested that 2 different animals were being measured. To resolve this, C-Lock Inc. received a cross-reference list at the start of each measurement period to replace the milking RFID with the ear tag number, ensuring accurate measurement assignment to individual animals.

Data

The final gas production values, expressed as grams per day, were calculated by C-Lock Inc. and we subse-

quently performed statistical analyses on the validated dataset. A total of 51,064 CH₄ and CO₂ measurements were available with a visit duration of >2 min, which is the minimum requirement for a valid GreenFeed visit (Harrison, 2025). Methane values ranged from 14 to 994 g/d, and CO₂ values ranged from 5,424 to 22,117 g/d. Additionally, 882 potential outlier measurements were identified by C-Lock Inc., with CH₄ ranging from 2 to 1,998 g/d and CO₂ ranging from 238 to 28,248 g/d. The outlier detection was performed separately for each day and GreenFeed unit. For CH₄, values exceeding the median by more than 2.5 SD were identified as outliers, and for CO₂, outliers were defined as values outside ± 2.5 SD from the median. C-Lock Inc. excludes CH₄ records only when values exceed the median and does not remove low values, which is particularly important when CH₄ reducing feed additives are used. Because GreenFeed measurements are subject to short visit durations and breath-to-breath variability, which can occasionally produce low but technically valid emissions, we also retained these low values in our analysis. After review, all potential outliers that fell within the range of the validated dataset were retained. As a result, only 51 measurements were excluded as outliers, resulting in a final dataset of 51,895 measurements from 451 animals from 9 commercial farms.

Statistical Analyses

Data processing, visualization, and statistical analyses were performed using R Statistical Software (v4.2.3; R Core Team, 2023).

Visit Duration

It is generally recommended to include only GreenFeed measurements with a visit duration of at least 3 min (Huhtanen et al., 2015). However, to avoid losing valuable data, we tested whether excluding records with a visit duration of less than 3 min was justified. An ANOVA was applied to assess differences in CH₄ and CO₂ emissions across 5 visit duration categories: <3, 3, 4, 5, and ≥6 min. To quantify the magnitude of the observed differences, eta-squared (η^2) was calculated, providing an estimate of the proportion of variance in emissions explained by differences in visit duration.

Variability

To determine the magnitude of diurnal, day-to-day, and within-cow variability in CH₄ and CO₂ emissions, the CV was calculated. The CV is calculated as the ratio of the SD to the mean (μ), expressed as a percentage. Diurnal and day-to-day variability were determined separately for each farm and measurement period (period 1 and period 2) because variability may differ between periods due to changes in feeding or management. Day-to-day variability was calculated only for herd-days with at least 10 GreenFeed visits. This threshold was applied to exclude variability associated with installation or deinstallation days, as well as days affected by technical issues. Within-cow variability was also calculated separately for each measurement period (period 1 and period 2).

Minimum Number of Required Spot Samples

Only animals with at least 70 measurements in either period 1 or period 2 were included in the analysis, resulting in 400 CH₄ and CO₂ records from 298 animals that met this criterion. Two approaches were used to evaluate the minimum number of GreenFeed spot samples required for reliable CH₄ and CO₂ estimates. In the first approach (forward sampling), subsets of the first 1, 10, 20, 30, 40, 50, 60, and 70 measurements per cow were compared with the cow's full set of available measurements, following the method of Dressler et al. (2023). In the second approach (random sampling), the same number of spot samples was selected randomly from all available visits for each cow, and this procedure was repeated 100 times to account for sampling variability. Random sampling was previously applied by Koning et al. (2024).

For both approaches, the Pearson correlation (r), the coefficient of determination (R^2) and the root mean square error (RMSE) between the arithmetic mean of the subset and that of all available measurements were calculated to evaluate accuracy. The minimum number of spot samples was defined as the number at which r reached a value of 0.90. In addition, for forward sampling, the mean number of days required to obtain the respective number of spot samples was calculated.

RESULTS AND DISCUSSION

GreenFeed Visiting Behavior

On average, cows visited the GreenFeed units for 4.25 min, with durations ranging from >2 min to 17 min. Analysis of variance results indicated that visit duration significantly affected mean CH₄ and CO₂ emissions ($P < 0.001$). However, the effect sizes were small ($\eta^2 = 0.016$ for CH₄ and $\eta^2 = 0.011$ for CO₂), as is also illustrated by the density plots (Figure 2). Longer measurement periods (≥3 min) are generally recommended to capture 3 or more eructations per visit, providing a representative estimate of CH₄ emissions (Huhtanen et al., 2015). In our study, however, to retain as many records as possible and given the small differences observed, all spot samples were retained for further analysis.

The average number of visits per cow was 67.1 in period 1 and 71.6 in period 2, with a minimum of 1 and a maximum of 215 visits, and an average of 2.6 visits/d. The highest frequency of visits was observed between midnight and 0100 h, followed by a decrease in activity. The lowest frequency of visits was between 0400 and 0500 h. After this, visit frequency increased and remained relatively stable throughout the day, with 3 time periods showing lower visit frequencies, late morning, early afternoon and late evening (Figure 3). The GreenFeed visiting behavior observed for each individual farm closely mirrored the overall pattern (results not shown). A similar GreenFeed visiting behavior was observed by Della Rosa et al. (2025) in dairy cows on a research farm in New Zealand. The frequency of visits decreased between midnight and 0500 h. It was highest in the period between morning and afternoon milking and after afternoon milking until 1800 h. After that, the visit frequency decreased during the next 2 h and then gradually increased again until midnight.

Emission Measurements

Mean CH₄ and CO₂ emissions were 430 g/d and 13,309 g/d, respectively. Differences in CH₄ and CO₂ emissions were observed both between farms and between periods within the same farm (Table 2). Overall,

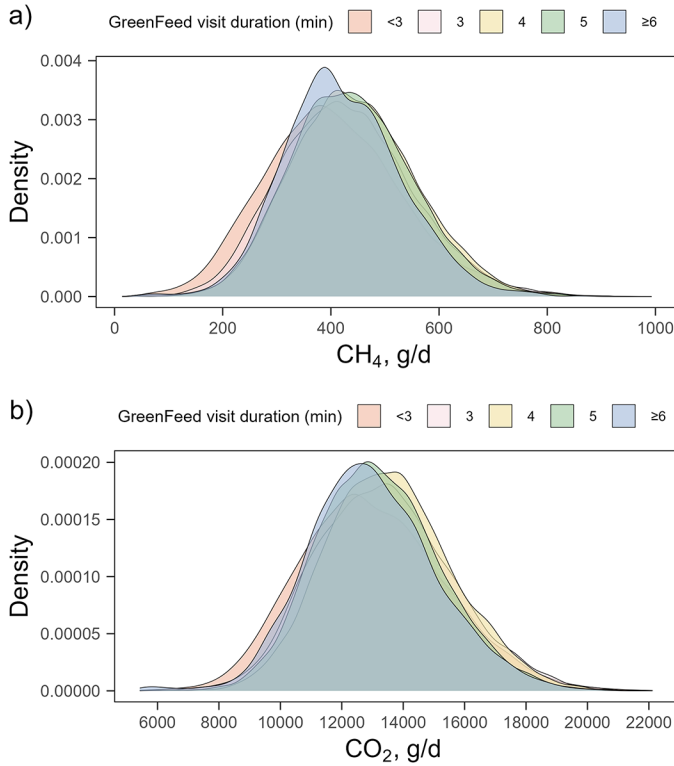


Figure 2. Density plots of (a) CH₄ and (b) CO₂ emission measurements by duration of GreenFeed visit.

the emission values were comparable to those reported for other dairy breeds measured using GreenFeed. For example, van Breukelen et al. (2023) reported mean emissions of 436 g/d for CH₄ and 13,159 g/d for CO₂ in lactating Holstein cows across 16 commercial farms in the Netherlands. Similar CH₄ emission values have been documented in other breeds, including 427 g/d in 194 Swedish Red cows from a single herd (Chipondoro, 2024) and 422 g/d in Norwegian Red cows (Heringstad et al., 2023).

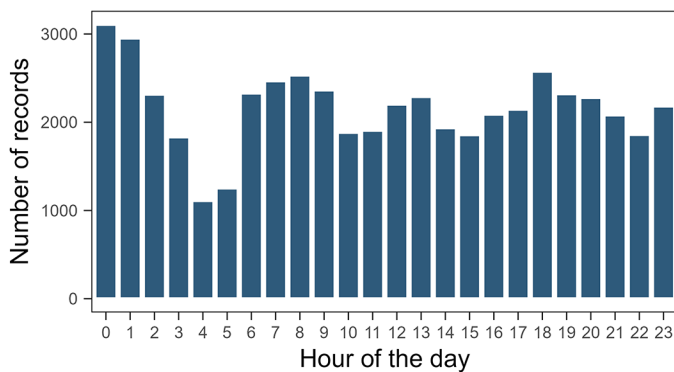


Figure 3. Number of GreenFeed visits (records) by hour of the day across all farms.

Table 2. Number of GreenFeed visits, number of cows, mean values for CH₄ and CO₂, and CV for diurnal and day-to-day variability in CH₄ and CO₂ for each farm and period

Item	Period	Farm									
		All	A	B	C	D	E	F	G	H	I
Number of GreenFeed visits	1	51,895	3,970	3,538	2,054	2,723	4,007	1,540	1,428	3,207	2,713
	2		2,899	3,360	3,459	3,466	2,721	2,299	1,595	4,457	2,459
Number of cows	1	451	55	55	38	35	49	19	24	55	45
	2		51	52	40	34	41	27	29	55	42
Mean CH ₄ (g/d)	1	430	409	456	482	440	403	496	381	425	416
	2		397	387	500	442	444	452	374	419	447
Mean CO ₂ (g/d)	1	13,309	13,733	14,109	12,314	13,717	13,367	14,051	11,813	12,651	13,074
	2		13,584	12,844	14,061	13,959	14,420	13,536	11,717	11,904	13,747
Diurnal CV of CH ₄	1		4.6	6.1	6.2	10.3	7.2	6.0	7.6	8.8	5.0
	2		5.3	8.2	6.1	12.0	4.6	5.8	9.8	8.2	8.0
Diurnal CV of CO ₂	1		3.3	4.5	3.9	6.5	4.5	3.6	4.2	5.6	3.9
	2		2.7	4.4	3.4	6.9	2.7	3.5	4.9	5.3	5.5
Day-to-day ¹ CV of CH ₄	1		5.6	4.6	3.7	5.0	5.3	6.4	5.6	5.5	4.3
	2		4.8	8.7	5.3	5.2	4.1	5.3	5.2	4.3	6.0
Day-to-day ¹ CV of CO ₂	1		5.0	5.5	4.0	3.3	2.5	3.2	3.8	3.2	3.5
	2		5.2	4.0	3.9	3.4	3.2	3.6	2.8	3.3	3.6

¹Only days with at least 10 measurements were considered in the analysis.

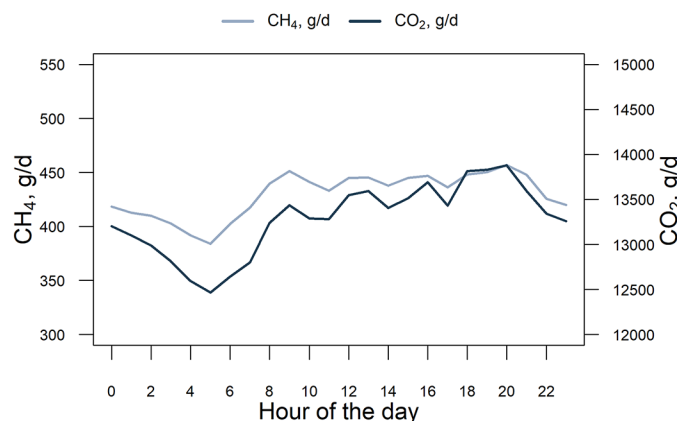


Figure 4. Diurnal variability in CH₄ and CO₂ emissions across all farms.

Variability in Emission Measurements

Diurnal Variability. Figure 4 shows the diurnal variability in CH₄ and CO₂ emissions across all farms. Methane and CO₂ emissions were lowest at 0500 h, with average values of 384 g/d for CH₄ and 12,468 g/d for CO₂. Emissions then increased and remained at a high level until 2000 h, after which they began to decline again. Diurnal patterns of CH₄ have been well-documented (Manafiazar et al., 2017; Islam et al., 2023; Raynor et al., 2025), with emissions typically higher after feeding, particularly in the morning and evening, when the rumen is most active. In contrast, emissions tend to decrease at night when cows are resting, and their digestive and metabolic processes slow down. A similar pattern was observed in the study by Fresco et al. (2023), where CH₄ emissions increased from 0900 (overall minimum) to 1100 h (overall maximum), remained stable until 2100 h, then decreased for the following 12 h. van Breukelen et al. (2023) reported that CH₄ emissions decreased during the night and were lowest around 0600 h, whereafter the measured CH₄ emissions increased. The CV for diurnal variability for each farm and period is presented in Table 2. Coefficient of variation values ranged from 4.6% to 12.0% for CH₄ and from 2.7% to 6.9% for CO₂, indicating low to moderate diurnal variability.

Day-to-Day Variability. Figure 5 shows the day-to-day variability in CH₄ emissions, displaying all farms separately for period 1 and period 2. The CV for day-to-day variability in CH₄ and CO₂ emissions for each farm and period are summarized in Table 2. The CV values ranged from 3.7% to 8.7% for CH₄ and from 2.5% to 5.5% for CO₂, indicating low day-to-day variability in emissions across the participating farms. The low day-to-day variability suggests consistent emission levels due to stable feeding systems and management practices. All participating farms were characterized by medium-to-high

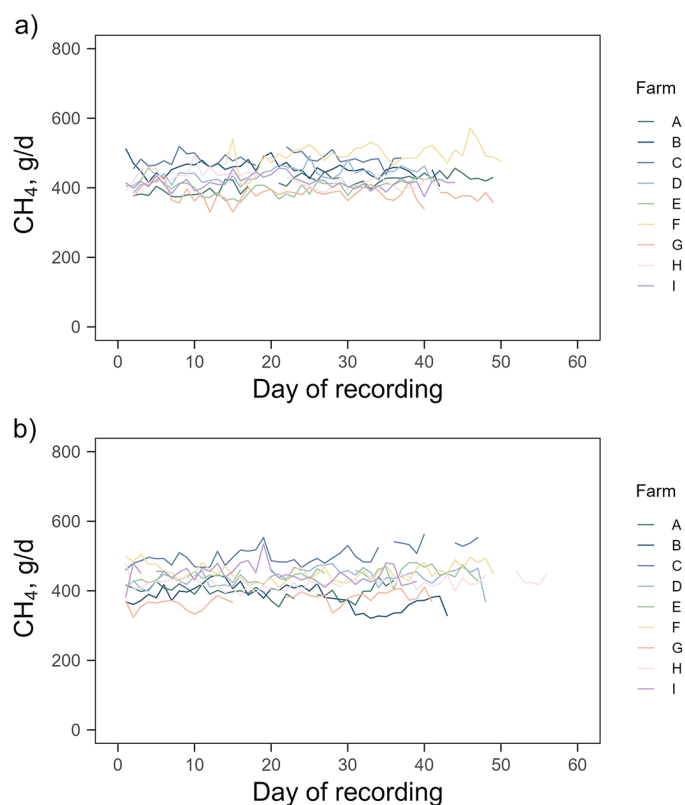


Figure 5. Day-to-day variability in CH₄ emissions for each farm, including only days with at least 10 measurements in (a) period 1 and (b) period 2.

milk performance and a high level of automation. This included automated milking systems or milking parlors with automatic milk meters and sensor technology. The cows were fed either a TMR or a partial mixed ration. These consistent and controlled feeding practices probably contributed to the low day-to-day variability.

Within-Cow Variability. Significant variability in CH₄ and CO₂ emissions was observed across repeated visits of the same cows. Figure 6 illustrates the within-cow variability of CH₄ emissions for all cows with GreenFeed measurements in period 1 and period 2. The mean within-cow CV for CH₄ was 22.2% in period 1 and 22.9% in period 2. For CO₂, the corresponding values were 11.8% and 11.5% in period 1 and period 2, respectively. This variability highlights the dynamic nature of CH₄ production, whereas CO₂ emissions are much more stable. As noted by Hegarty (2013), CH₄ emissions from individual animals are not constant throughout the day and depend at any given time on (1) the time relative to feeding, the amount of feed, and feed intake patterns, (2) the composition of the feed, and (3) short-term fluctuations in gas release from the rumen. A recent study by de Mol et al. (2024) showed that CH₄ emissions measured using the GreenFeed were significantly influenced by the amount

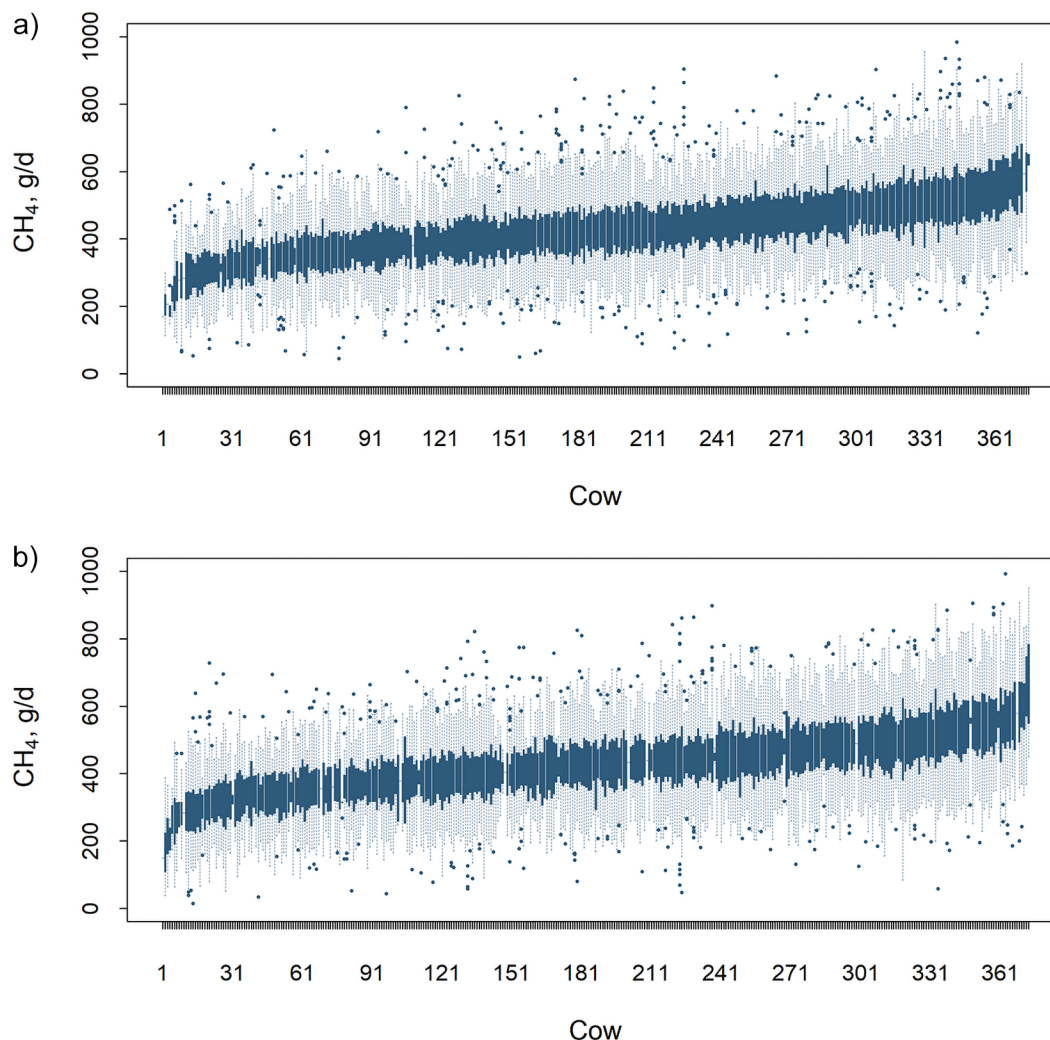


Figure 6. Within-cow variability in CH₄ emissions: boxplots of spot samples per cow, sorted in ascending order by mean CH₄ emissions: (a) period 1 (N = 375 cows) and (b) period 2 (N = 373 cows).

of feed consumed during the previous meal and the time interval between the last meal and the GreenFeed visit. However, these factors could not be accounted for in our study because the necessary data were not recorded on the farms. By determining the minimum number of spot samples required, the current study helps to reduce potential bias due to variations in feeding behavior and ensures that the measurements reflect the actual differences in emissions.

Minimum Number of Required Spot Samples

Table 3 shows the different number of spot samples and their effect on CH₄ and CO₂ emission estimates based on r , R^2 , and RMSE, and the mean days required to obtain them using forward sampling. For CH₄, as the number of spot samples increased from 1 to 70, r improved from

0.49 to 0.99, R^2 increased from 0.24 to 0.97, and RMSE decreased from 110 to 12. Based on these results, at least 19 spot measurements are required to achieve acceptable accuracy, corresponding to r of 0.90, R^2 of 0.81 and RMSE of 33. The mean days required to obtain them was 6.5 d. Figure 7 shows the distribution of the number of days required for cows to reach 19 spot samples. In contrast, for CO₂, a similar level of accuracy can be achieved with fewer measurements; only 13 spot samples were sufficient to reach r of 0.90, R^2 of 0.82, and RMSE of 720. The mean days required to obtain them is 4.3 d. The guidelines paper by Jonker et al. (2020) also states that a larger number of samples is typically required to accurately quantify CH₄ emissions compared with other respiratory gases. This is because CH₄ is primarily emitted through eructation pulses that occur only about once every 45 to 60 s, whereas CO₂ is emitted through normal

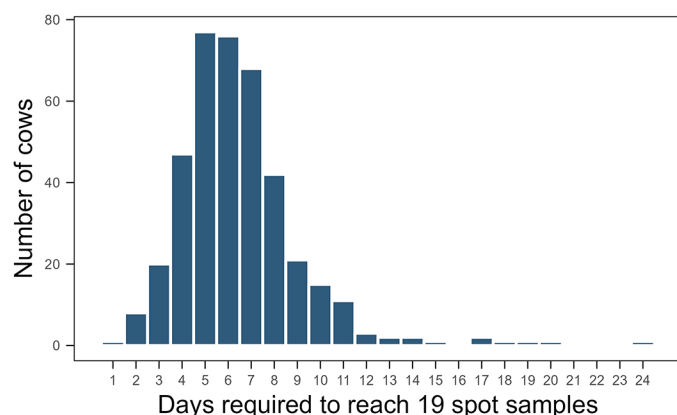
Table 3. Forward sampling: number of spot samples required for accurate CH₄ and CO₂ emission estimates based on *r*, R², and RMSE, and the mean days required to obtain them¹

Number of spot samples	Mean days	CH ₄			CO ₂		
		<i>r</i>	R ²	RMSE	<i>r</i>	R ²	RMSE
1	—	0.49	0.24	110	0.58	0.34	1,822
10	3.3	0.84	0.70	43	0.89	0.79	779
13	4.3	0.86	0.74	40	0.90	0.82	720
19	6.5	0.90	0.81	33	0.93	0.86	608
20	6.9	0.90	0.82	32	0.93	0.87	596
30	10.6	0.93	0.87	26	0.95	0.90	505
40	14.4	0.95	0.90	22	0.96	0.93	427
50	18.4	0.97	0.94	18	0.97	0.95	356
60	22.4	0.98	0.96	15	0.98	0.96	296
70	26.3	0.99	0.97	12	0.99	0.98	232

¹Bold entries indicate the number of spot samples at which *r* reaches 0.90.

respiration, making its emissions less variable (Huhtanen et al., 2015). In a recent study, Dressler et al. (2023) found that, based on forward sampling, a minimum of 38 and 40 spot samples were required for the quantification of CH₄ and CO₂, respectively, in grazing beef cows. However, in their study, the minimum number of spot samples was determined when correlations with the full 100 visits were greater than 0.95. Applying the same threshold in our study requires 40 spot samples for CH₄ (mean 14.4 d) and 31 spot samples for CO₂ (mean 10.9 d).

In contrast, when using random sampling (Table 4), only 8 spot samples for CH₄ and 5 spot samples for CO₂ were sufficient to meet the accuracy criteria, which is substantially lower than the number required with forward sampling. Similarly, Koning et al. (2024), using a random sampling approach, found that a minimum of 17, 10, and 11 spot samples were required for reliable CH₄ emission estimates in dairy cattle fed fresh grass outdoors (grazing), fresh grass indoors, and grass silage indoors, respectively.

**Figure 7.** Distribution of the number of days required for cows to reach 19 spot samples.**Table 4.** Random sampling: number of spot samples required for accurate CH₄ and CO₂ emission estimates based on *r*, R², and RMSE¹

Number of spot samples	CH ₄			CO ₂		
	<i>r</i>	R ²	RMSE	<i>r</i>	R ²	RMSE
1	0.57	0.33	97	0.69	0.48	1,548
5	0.85	0.72	42	0.91	0.83	678
8	0.90	0.81	33	0.94	0.89	524
10	0.92	0.85	29	0.95	0.91	466
20	0.96	0.93	19	0.98	0.96	311
30	0.98	0.96	15	0.99	0.98	238
40	0.99	0.97	12	0.99	0.98	189
50	0.99	0.98	9	0.99	0.99	153
60	0.99	0.99	8	0.99	0.99	124
70	0.99	0.99	6	0.99	0.99	99

¹Bold entries indicate the number of spot samples at which *r* reaches 0.90.

Our results showed that variability in reported sample requirements across studies is the result of, among other things, different accuracy thresholds and methodological approaches. As noted by Jonker et al. (2020), the number of spot samples required depends on the intended application. For studies requiring high precision, such as controlled feeding experiments or treatment comparisons, a larger number of samples is often essential (MacNeil and Waterman, 2025). In cases where an observed response is tested against a threshold, statistical power becomes particularly important, for example, if tax policies are based on achieving specific levels of GHG emissions (Bistline et al., 2024) or when validating predictive models of GHG emissions (Gavrilova et al., 2019). In contrast, for genetic studies, which is the focus of the breed4green project, fewer spot samples per animal may be used to enable analysis of a larger number of cows, thereby improving the accuracy of heritability and genetic correlation estimates.

Different methods lead to different recommendations. Forward sampling, which is based on the actual sequence of visits made by each cow to the GreenFeed, offers a more conservative and realistic approach compared with random sampling, although it also comes with certain limitations. In the present study, each period lasted approximately 6 wk, and this time frame was used as the observation period to evaluate how many samples are required to achieve a certain level of accuracy. Because the commercial farms fed the cows the same diet with a constant feeding strategy, this was not expected to be a problem. Forward sampling assumes that emissions measured during the first week are representative of those in later weeks. Although this assumption may hold when cows are fed the same diet, as in the present study, cows advance in DIM, which is associated with changes in DMI and shifts in energy balance, and these factors in turn influence emissions. However, over the 6-wk period, the cows were still within the same lactation stage (early, mid or late lactation), which likely

minimized the effects of lactation stage on the results. Consequently, for future analyses on this dataset, we will rely on the results from forward sampling, which indicate that at least 13 spot samples per animal are required for CO₂ and at least 19 spot samples per animal are needed for CH₄. Furthermore, in future analyses, we will evaluate the effect of applying stricter accuracy thresholds, as proposed by Dressler et al. (2023). In doing so, the trade-off between increased data accuracy and retaining a sufficient number of animals for robust and meaningful analyses will be explicitly considered.

A key challenge when using commercial farm data is that not all cows meet the required minimum number of measurements for a robust estimation of CH₄ and CO₂. In this study, only 298 out of 451 cows (64% of all cows) met the requirement of at least 70 spot samples in either period 1 or period 2, which may affect the representativeness of the dataset. In addition, subset means were compared with the mean calculated from the complete dataset, implicitly assuming that the latter represents each cow's true emission value. When the total number of observations for a cow is close to the subset size (e.g., 70 out of 75 records), the subset mean will inevitably be highly similar to the full dataset mean, resulting in inflated *r* and *R*² values. In contrast, cows with a much larger number of total observations (e.g., 100 records or more) are less likely to achieve similarly high *r* and *R*² values for the same subset size. This introduces an inherent bias related to the total number of observations per cow, which should be considered when interpreting the results.

Overall, we recommend checking the minimum sample size needed for each study on commercial farms based on our analysis, as this ensures that the sample number is optimized for the specific conditions of the study.

CONCLUSIONS

A minimum of 19 spot samples were required for CH₄ estimates, and 13 were sufficient for CO₂. The substantial variability observed in CH₄ emissions underscores the complex and dynamic nature of CH₄ production, influenced by factors such as feeding patterns and rumen dynamics. In contrast, CO₂ emissions showed greater stability, with fewer samples required to achieve accurate estimates. The minimum number of spot samples varies depending on the study design and the required accuracy. It is therefore essential to determine the minimum number of spot samples for each study on commercial farms before analysis.

NOTES

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Nonstandard abbreviations used: AMS = automatic milking system; BS = Brown Swiss; FV = Fleckvieh; RFID = radiofrequency identification; RMSE = root mean square error.

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